

Homework (Habanero)

- Q1.** A doctor prescribes 2 pills of 10mg each to a patient. What is the ratio of the total dosage to the number of pills?
- (A) 1:5
(B) 1:2
(C) 2:1
(D) 5:1
(E) 10:1
- Q2.** A cell divides into 3 parts, with the ratio of the resulting cells being 1 : 2 : 3. If the original cell had a volume of $30\mu m^3$, what is the volume of the smallest cell?
- (A) $5\mu m^3$
(B) $6\mu m^3$
(C) $7\mu m^3$
(D) $8\mu m^3$
(E) $10\mu m^3$
- Q3.** A medical trial has a success rate of 4 : 7. If 140 participants are enrolled, how many are expected to succeed?
- (A) 32
(B) 40
(C) 48
(D) 56
(E) 64
- Q4.** What is the value of x in the proportion $\frac{2}{3} = \frac{6}{x}$?
- (A) 6
(B) 8
(C) 9
(D) 12
(E) 18
- Q5.** A nurse is preparing a solution with a 3 : 5 ratio of medication to water. If she uses 15ml of medication, how much water should she add?
- (A) 20ml
(B) 25ml
(C) 30ml
(D) 35ml
(E) 40ml
- Q6.** The growth rate of a bacteria culture is 2 : 3. If the culture has 20 bacteria after 2 hours, how many bacteria will there be after 4 hours?
- (A) 40
(B) 50
(C) 60
(D) 80
(E) 100
- Q7.** What is the simplified ratio of 18 : 24
- (A) 1:2
(B) 2:3
(C) 3:4
(D) 4:5
(E) 5:6
- Q8.** A doctor prescribes x pills to a patient, with a ratio of 2 : 5 for the dosage. If the patient takes 10 pills, what is the dosage?
- (A) 4mg
(B) 6mg
(C) 8mg
(D) 10mg
(E) 12mg

Open Ended Questions (FRQ)

- Q9.** The ratio of boys to girls in a hospital is 7 : 9. If there are 20 more girls than boys, how many boys and girls are there in total? Show all work and calculations.
- Q10.** A new medicine has a 3 : 10 ratio of active ingredients to inactive ingredients. If a patient is prescribed 50mg of the active ingredient, how much of the inactive ingredient should they take? Explain your reasoning and calculations.

Chef's Tips

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- Tip 1: Ensure students use correct units and labels in their calculations.
- Tip 2: Encourage students to check their work and calculations for accuracy.
- Tip 3: Remind students to show all work and reasoning for free-response questions.